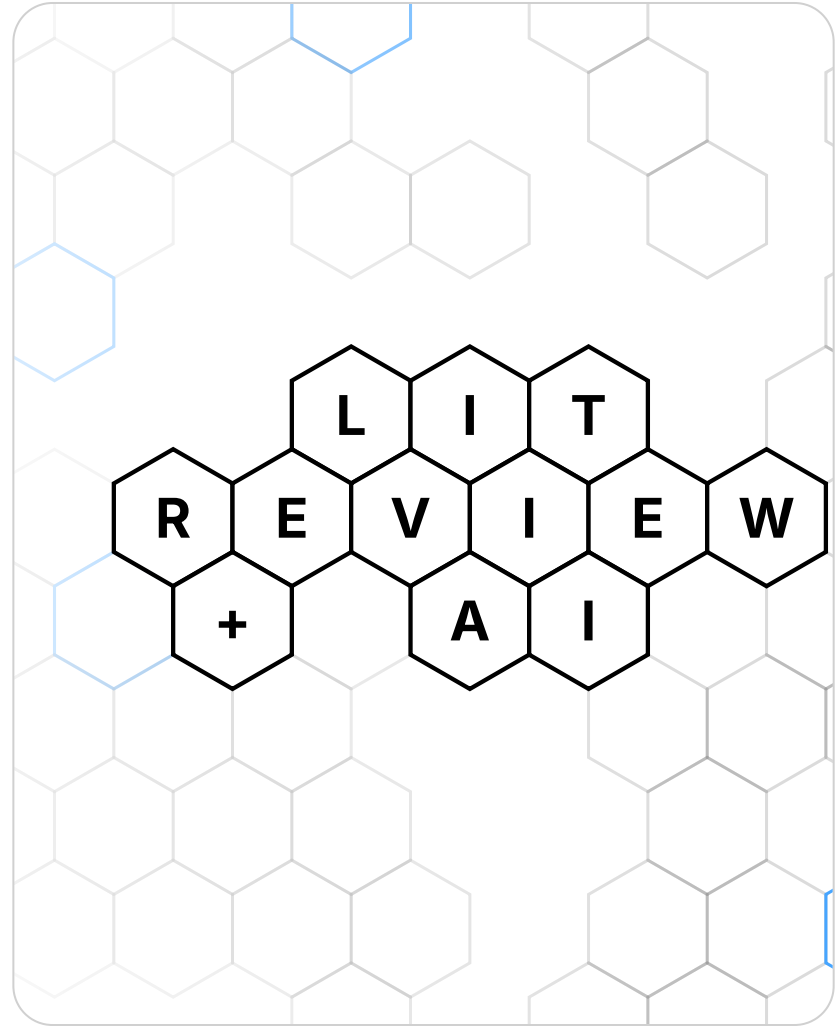
A!

Literature reviews and AI

A short introduction



Enrico Glerean, Staff Scientist, Aalto University, October 2026

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Outline

1. **Foundations of AI in Research**: what generative AI is, where it can help across the research lifecycle, the misconduct risks it creates, and when you're required to disclose using it
2. **Literature Review Foundations**: what a literature review is, the main review types, and the 12-step process behind a systematic one
3. **AI-Assisted Literature Review**: a step-by-step look at the evidence on how well AI tools actually perform each of those 12 steps
4. **A Framework for Responsible AI-Assisted Literature Review**: a five-level spectrum, from AI as consultant to AI as autonomous reviewer, for deciding how much of the process to delegate

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AI in research work



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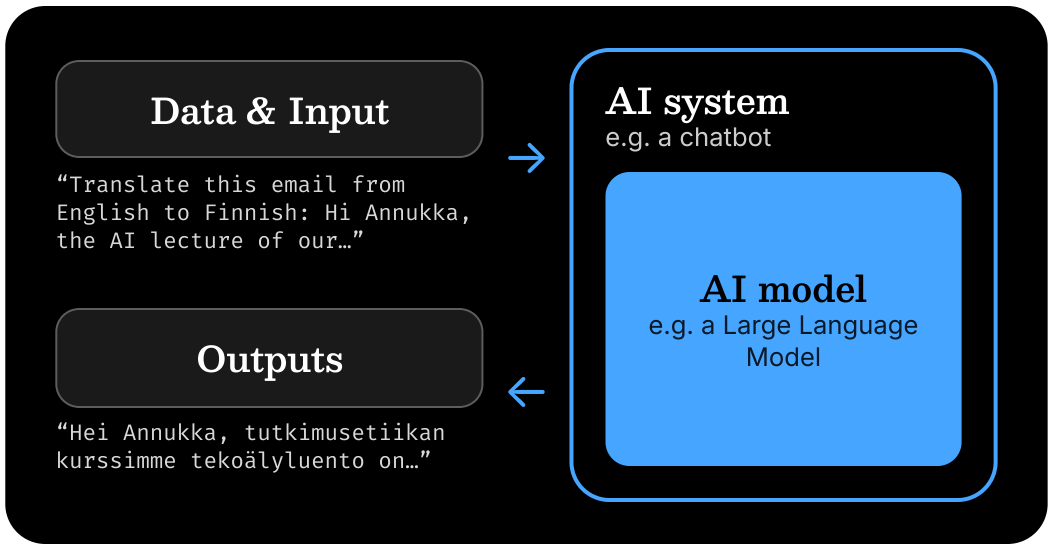
What is an AI system?

Figure adapted from Glerean (2025) "Fundamentals of secure AI systems with personal data", European Data Protection Board

For simplicity we focus on **generative AI systems** that use **General Purpose AI models** such as **Large Language Models** that can be used in your research work, including **agentic AI** systems.

Generating one modality from another

- Text → Text
- Text → Code
- Speech → Text
- Text → Speech
- Text → Image
- Image → Text
- ...



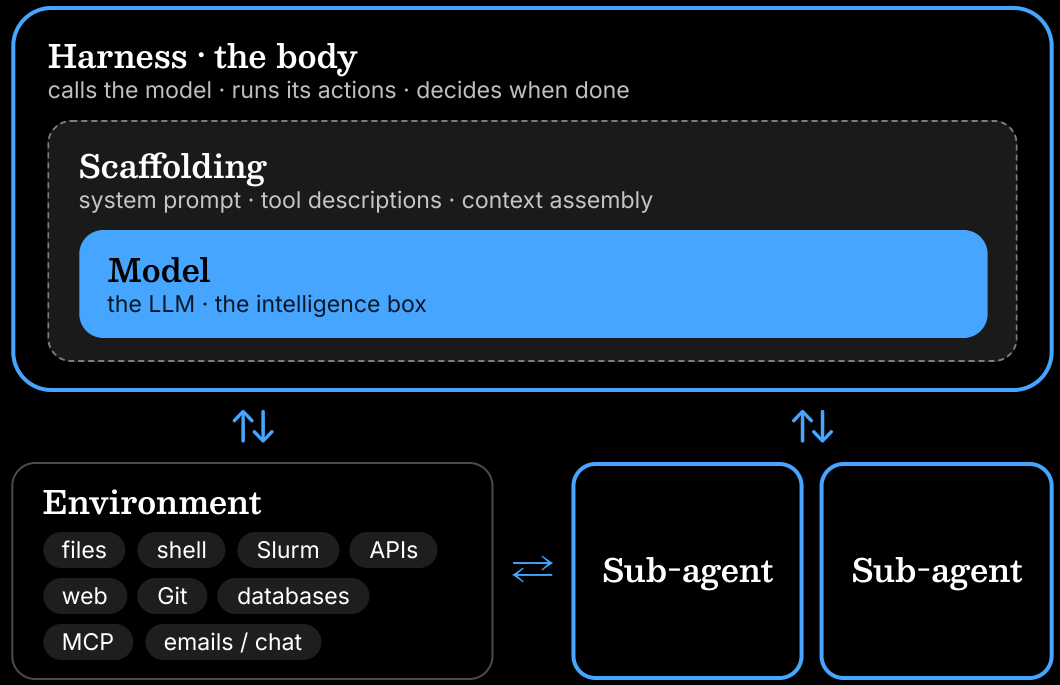
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What is an AI agent?

The four elements of an AI agent: **a) goal complexity, b) environmental complexity, c) adaptability, d) independent execution.** (OECD 2026).

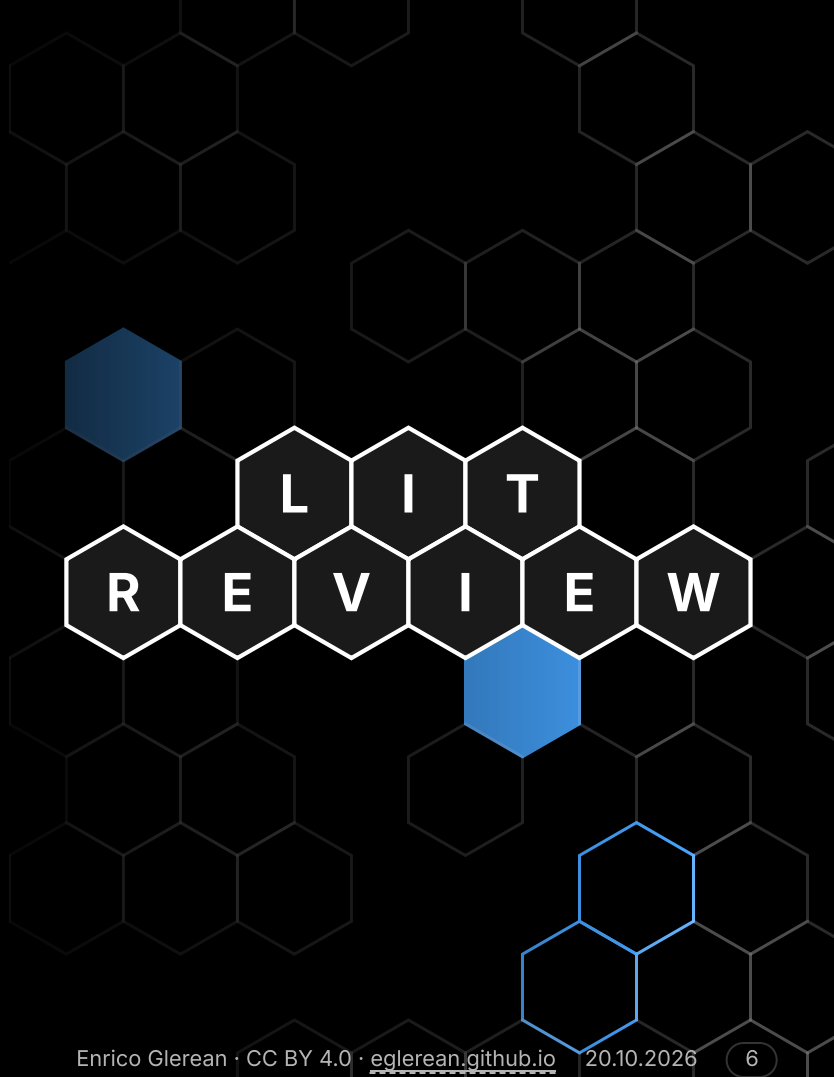
An AI agent pairs the model with more components so that it can **take real actions**.

- **The model**, the "intelligence box": the LLM itself.
- **The scaffolding**, between the two: system prompts, tool descriptions, and the rules for how context is assembled and formatted before it reaches the model.
- **The harness**, the "body": the code around the model that calls it, carries out whatever it asks for, and decides when the task is done.



Adapted from Hugging Face, "Harness, Scaffold, and the AI Agent Terms Worth Getting Right", 2026

Literature Review Foundations



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What is a literature review?

Grant & Booth (2009) · Useful reference: "Systematic searching", University of Oulu

A literature review **identifies, evaluates and synthesises** existing research on a topic.

How explicit and reproducible each of these steps is defines the **type** of review: from a narrative review, where the author chooses and interprets the sources, to a systematic review, where every step is planned in advance and reported so that others can repeat it.

THE 14 REVIEW TYPES OF GRANT & BOOTH (2009)

Critical review Evaluates, then builds new concepts	**Literature review** Generic summary of published work	**Mapping review / systematic map** Maps and categorises existing literature	**Meta-analysis** Statistically combines quantitative results	**Mixed studies review** Combines quantitative and qualitative studies	**Overview** Broad survey of a topic	**Qualitative systematic review** Integrates findings of qualitative studies
Rapid review Systematic methods, shortened timeline	**Scoping review** Size and scope of literature	**State-of-the-art review** Most current research only	**Systematic review** Exhaustive search, appraisal, synthesis	**Systematic search and review** Comprehensive search, best-evidence synthesis	**Systematized review** Some systematic steps, not all	**Umbrella review** Synthesises reviews, not studies

Broadly, they form **three families**: **systematic**, **narrative** and **meta-analysis / meta-synthesis**.

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Three families of literature reviews

1. Systematic review

The most rigorous form: every step is planned in advance and reported.

- Answers a **well-defined research question**
- The search is carried out **systematically** and **reported accurately**
- **Inclusion and exclusion criteria** are defined before screening starts
- The **methodological quality** of the selected studies and their **risk of bias** are carefully assessed
- Related variants include the umbrella review, the integrative review, and the systematized review

2. Narrative review

Sometimes just called a "literature review": the methodologically lightest form.

- Describes previous research on a topic, often addressing a **broad question**
- The literature search is **not carried out in a fully systematic way**
- Selective collection of studies can lead to **divergent descriptions of the same topic** depending on who wrote the review
- Sub-types include the critical review, the scoping review, the mapping review, and the overview — some of which borrow features from systematic reviews

3. Meta-analysis / meta-synthesis

Combines results across studies rather than just describing them.

- A **quantitative meta-analysis** statistically combines and generalizes numerical results from multiple studies
- A **qualitative meta-synthesis** (or meta-summary) classifies and summarizes findings thematically instead of numerically

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Review types

Grant & Booth (2009) | Sutton et al. (2019) | Munn et al. (2018)

Purpose	Review types	Method	Reporting / quality
Interpret a topic	Narrative, critical review	The author selects and interprets sources; quality depends on being transparent about how sources were chosen	SANRA
Map what exists	Scoping, mapping review	Systematic search that charts the evidence; usually no quality appraisal	PRISMA-ScR
Answer a focused question	Systematic review, with or without meta-analysis, or qualitative evidence synthesis	Registered protocol, systematic search, appraisal, synthesis	PRISMA 2020
Synthesise reviews	Umbrella review	A systematic review of systematic reviews	—
Answer fast	Rapid review	Systematic methods with simplified steps: completeness traded for speed	—

Meta-analysis is a statistical step inside a systematic review, not a review type of its own. Scoping reviews differ from systematic reviews in purpose, not rigour. Sutton et al. (2019) describe 48 review types in seven families.

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The 13 steps of a systematic review

Synthesised from the Cochrane Handbook and Kolaski et al. (2023); see also Tsafnat et al. (2014); item numbers from PRISMA 2020 checklist

#	Step	What to do	PRISMA 2020
1	Question & review type	Frame the question, pick review type (PICO, SPIDER)	3–5
2	Check existing reviews	Search registries: PROSPERO (medical), OSF (other)	3
3	Protocol & registration	Pre-register your plan (PRISMA-P)	24
4	Search strategy	Turn concepts into search terms (Boolean syntax, MeSH, Emtree) and free-text synonyms (PRESS)	7
5	Database searches	Run the search in several databases; deduplicate	6, 16a
6	Supplementary searching	Citation chaining (backward and forward), experts, grey literature, preprints (PRISMA-S)	6–7
7	Title/abstract screening	Independent screeners against the eligibility criteria	8, 16a
8	Full-text screening	Record the reason for every exclusion	8, 16b
9	Data collection	Pull study data into data items	9–10
10	Quality appraisal	Judge risk of bias: RoB 2 (trials), ROBINS-I (non-randomised studies), JBI/CASP checklists	11, 18
11	Synthesis & certainty	Meta-analysis or synthesis without meta-analysis (SWiM); rate certainty of evidence (GRADE)	12–15, 19–22
12	Reporting	PRISMA 2020 checklist and flow diagram (PRISMA-ScR for scoping reviews)	all
13	Updating	Re-run and update; living reviews (PRISMA-LSR)	PRISMA-LSR

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Not every field works like the health sciences

Kitchenham et al. (2009) | Parida & Brown (2018)

The 13 steps come from **evidence-based medicine**, where review methods are the most formalised. They are the **reference standard**, also for engineers and economists.

Other fields keep the logic but **change the tools**: their own databases and vocabularies, their own grey literature, their own appraisal concerns. Software engineering, for example, wrote its own guidelines.

Learn the medical model, then your field's guidance. More interpretive traditions (design, humanities) combine, reorder or skip some steps.

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Method guidance and databases by field

Fields: OECD Fields of Research and Development, Frascati Manual (2015)

Field (OECD)	Method guidance	Most used databases
Natural sciences (maths, computer and information sciences, physics, chemistry, earth, environmental and biological sciences)	CEE Guidelines and ROSES (environment); PRISMA-EcoEvo (ecology and evolution)	Web of Science, Scopus; field databases: GeoRef, Inspec, CAS SciFinder, MathSciNet/zbMATH; arXiv preprints
Engineering and technology (electrical, mechanical, civil, chemical, materials, biomedical, environmental engineering)	Kitchenham et al. 2009 (software engineering, the usual template also for engineering); no general engineering standard	Compendex (Engineering Village), Inspec, IEEE Xplore, ACM Digital Library, Scopus
Medical and health sciences	Cochrane Handbook; JBI Manual; PRISMA 2020	PubMed/MEDLINE (MeSH), Embase (Emtree), Cochrane CENTRAL, CINAHL, PsycINFO
Agricultural and veterinary sciences	CEE Guidelines; SYRCLE risk of bias (animal studies)	CAB Abstracts, AGRICOLA, AGRIS, Web of Science
Social sciences (psychology, economics, education, sociology, law, political science)	Campbell Collaboration; MAER-Net guidelines (economics)	PsycINFO, ERIC, EconLit, Sociological Abstracts; RePEc, SSRN, NBER working papers
Humanities and the arts (history, languages, philosophy, arts)	No formal standard: mostly narrative and critical reviews	MLA International Bibliography, Philosopher's Index, Historical Abstracts, JSTOR

Across all fields: Scopus and Web of Science; Google Scholar only as a supplementary source (Gusenbauer & Haddaway 2020). Outside health, register protocols on **OSF Registries**: PROSPERO accepts only health-related reviews.

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AI assisted literature review

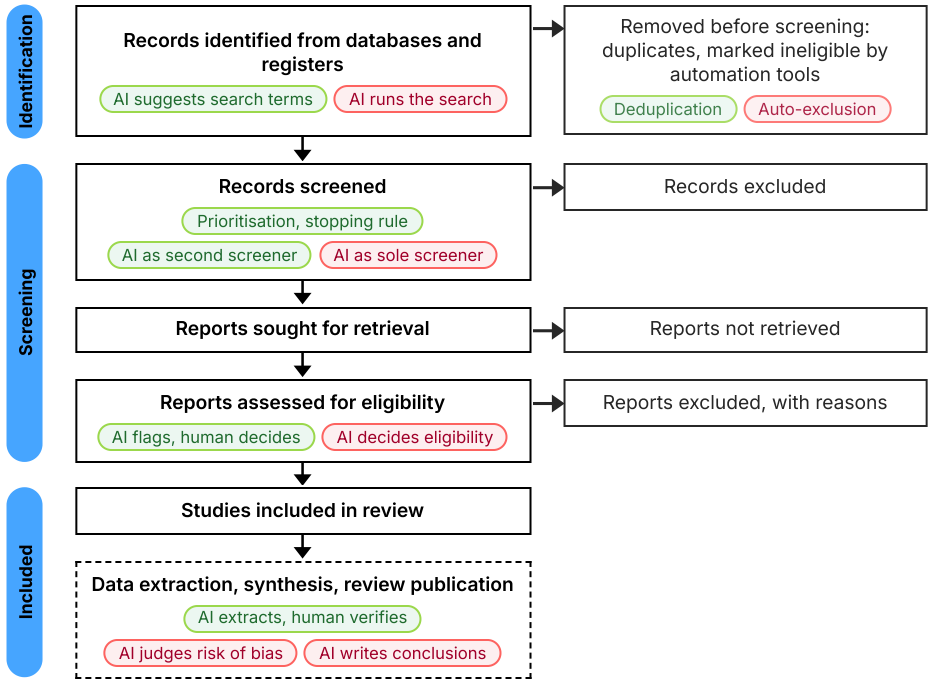


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Can we just add AI to each step?

Cochrane, Campbell, JBI and CEE position statement on AI use in evidence synthesis (2025)

- Each step can be **delegated** to an AI system or done **together** with one
- The more autonomy is left to the AI tool, the higher the **risk of undetected error**. Unchecked errors that reach a publication can become misconduct
- Use AI only where you can **show that it does not compromise the rigour** of the review. **The authors stay responsible** for everything in it



Flow after PRISMA 2020, as adapted by van Dijk et al. (2023)

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What tools are out there?

"Deep-research" modes (generative AI)

The agentic research features built into general chatbots (e.g. ChatGPT Deep Research, Gemini Deep Research, Claude in agent mode). Given a topic, these plan a search, browse the open web, and synthesize an answer with citations, **entirely on their own initiative**.

Classical machine learning for screening

ASReview (active learning), Rayyan and Covidence can **reorder records** so that relevant ones come first. It has been shown that this type of text mining saved **30–70% of screening workload**, sometimes at the cost of about 5% of relevant studies (O'Mara-Eves et al. 2015; van de Schoot et al. 2021). In one review with ASReview, the reviewers screened only **23% of the records** before reaching their stopping criterion (van Dijk et al. 2023).

LLM-based review assistants

Elicit and similar tools search a fixed corpus (Semantic Scholar for Elicit) and use LLMs to screen, summarise and extract. In four case studies Elicit found **25–69% of the studies** found by the original searches (Lau & Golder 2025).

Examples where AI can help

Draft a Boolean query

Test it against 5–10 papers you already know are relevant, or have a librarian review it. LLM-drafted queries are precise, but can also miss many relevant studies (Wang et al. 2023).

"Translate this research question into a Boolean search string using MeSH terms and free-text synonyms, formatted for PubMed: [question]."

Use AI as a second screener

In your prompt, you can ask the chatbot to answers with just "include / exclude / unsure" and report its justification. This can be compared with what other human screeners did.

"Eligibility criteria: [paste them]. For this title and abstract: [paste it] answer include / exclude / unsure, and quote the exact phrase that justifies your answer."

Extract data from studies

96.3% accuracy on simple trial data (Gartlehner et al. 2024), but a median of 14% errors across studies (Clark et al. 2025).

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Examples where AI fails

Running the search

In three evaluations, generative AI missed **68–96% (median 91%)** of the studies found by conventional searches (Clark et al. 2025). In general, some publishers/companies might block ChatGPT/Claude/Gemini search requests, making some part of the internet invisible to AI tools.

Being the only screener

Incorrect exclusions: median **28%** (range 1–83%) (Clark et al. 2025).

Citing literature from memory

29–91% of chatbot-suggested references were fabricated (Chelli et al. 2024). Note that this is an old study, things have gotten better, but still it depends on memorization or the availability of **web search** tools.

Summarising findings

LLM summaries were about **five times more likely** than expert summaries to overgeneralise (OR 4.85; Peters & Chin-Yee 2025).

Disclosing AI use in a review

Current best practices

- Cochrane, Campbell, JBI, CEE (2025), endorsing RAISE: report AI use transparently and describe **how outputs were verified**
- PRISMA 2020 items 8, 9 and 11 already ask for details of **automation tools** in selection, data collection and risk of bias
- EC Living Guidelines: disclose the AI tools that substantially shaped the work
- Remember to check the **journal's own AI policy**, which may be stricter

What to report: the GAMER statement

- Which tools and versions, and how they were prompted
- What role each tool played
- How the output was verified
- Data privacy
- The impact on the conclusions

Reporting guideline in the EQUATOR library (BMJ Evidence-Based Medicine, 2025). PRISMA 2020 is still the current version; an AI update is in development. PRISMA-trAIce (2025) uses the PRISMA name but is not an endorsed extension.

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A Framework for Responsible AI-Assisted Literature Review



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AI autonomy in literature reviewing

Framework: Glerean (2026), CC BY 4.0. Stages after Parasuraman, Sheridan & Wickens (2000)

HUMAN-DRIVEN → AI-DRIVEN

Consult	Discover	Assist	Delegate	Autonomous
"Help me query"	*"Help me find"*	*"Help me process"*	*"Carry out the review"*	*"Review this topic"*
WORDING	**RETRIEVAL**	**RELEVANCE**	**EXTRACTION**	**INTERPRETATION → SYNTHESIS**
synonyms	what to find	what matters	data to extract	what it means
concepts	where to search	include / exclude	categorise	conclude
query terms	similar papers	prioritisation		
information acquisition	*information acquisition*	*information analysis*	*decision selection*	*action implementation*

Examples show which decision moves to AI at each level; any step can be done at any level

Human methodological control	HIGH	LOW
Need for transparency, verification and validation	LOW	HIGH

More AI ≠ better review.
Report any use of AI that makes or suggests judgments (Position statement on AI use in evidence synthesis, 2025).

Some exercises with Duck.ai and Gemini

Open the tools: duck.ai · gemini.google.com

Build a good search query

"I want to search the scientific literature about whether social media use affects sleep in elementary school children. Identify the main concepts, synonyms and related terms. Then propose a Boolean search query. Explain why you chose each term."

Then try a topic from your field, in your field's database: wearable ECG arrhythmia detection (PubMed) · LLM coding assistants and software defects (ACM Digital Library, arXiv) · battery degradation models (IEEE Xplore) · low-traffic neighbourhoods and air quality (Scopus) · minimum wage and employment (EconLit)

Test "deep research" modes

First use **metaprompting** (prompt engineer is the first job that AI replaced :)): ask the tool to build a prompt for you on a research topic. Then use that prompt to run the deep research.

"I want you to generate the prompt to be used in deep research mode for conducting a literature review on the topic of functional connectivity and fMRI. I am interested in the recent findings from the past 2 years, especially with focus on new methods or methodological aspects. Ask me a few questions if you need to clarify the scope before you generate the prompt."

Check five references the tool gives you: do they exist, and do they say what the tool claims?